

Data Dictionary — Data Flow Diagram (Level 1)

E-Commerce Platform | System Documentation

1. External Entities

Actors and systems that interact with the e-commerce platform from outside its boundary.

ID	Name	Description	Inputs to System	Outputs from System
E1	Customer	End user who browses, purchases, and reviews products.	Credentials, checkout data, reviews, activity events	Products, order status, recommendations, receipts
E2	Admin	Staff member managing inventory, orders, and platform settings.	Product data, category updates, order status changes	Analytics reports, dashboard metrics, alerts
E3	Payment Gateway	External payment processors: Pesapal, MTN MoMo, Airtel Money, Stripe, PayPal.	Transaction result, IPN / webhook callback	Payment initiation request (amount, reference, redirect URLs)
E4	AI Service	FastAPI microservice hosting collaborative-filter and content-based recommendation models.	Ranked recommendation IDs, similarity scores	User behaviour summary, product ID lists for scoring

2. Processes

The seven functional processes that transform data within the system.

ID	Name	Description	Key Inputs	Key Outputs
P1	User Authentication	Handles registration, login, JWT issuance, email verification, and password reset.	Credentials, reset token	JWT token, user profile, verification email
P2	Product Management	CRUD operations for products and categories; Cloudinary image upload; stock tracking.	Product data, category data, image files	Product catalogue, image URLs, category tree
P3	Order Processing	Validates cart, reserves stock, creates order, manages status transitions (pending → delivered).	Cart contents, shipping address, coupon code	Order confirmation, stock update, payment intent
P4	Payment Processing	Initiates payment with the appropriate gateway, handles IPN / webhook callbacks, updates order and payment records.	Payment request, gateway webhook payload	Redirect URL, payment record, updated order status
P5	Recommendation Engine	Retrieves personalised or trending recommendations from the AI service; falls back to DB trending query if AI service is unavailable.	User ID, product IDs from AI service	Ranked and enriched product list delivered to customer
P6	Review and Wishlist	Creates, updates, and deletes product reviews; manages wishlist additions/removals; recalculates product rating aggregates.	Review data, wishlist action (add/remove)	Persisted review, updated product averageRating, wishlist state
P7	Activity Tracking	Logs user behavioural events (views, searches, cart actions, purchases) for analytics and AI model training data.	Event type, product ID, metadata	Activity log entry written to D8; aggregated data forwarded to AI

				service
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3. Data Stores

Persistent storage locations used by the system processes.

ID	Name	Technology	Description	Key Fields
D1	Users	MongoDB	All registered user accounts including credentials and embedded shipping addresses.	_id, email, password (hashed), role, isVerified, addresses
D2	Products	MongoDB	Product catalogue with pricing, inventory, imagery, and computed rating aggregates.	_id, name, price, stock, status, category, images, averageRating
D3	Orders	MongoDB	Customer orders from placement through fulfilment, including line-item snapshots.	_id, user, items, status, paymentStatus, total, shippingAddress
D4	Payments	MongoDB	Payment transaction records from all gateways with raw gateway response for audit.	_id, order, method, status, transactionId, amount, currency, gatewayResponse
D5	Cart	MongoDB	Persisted shopping carts — one per user — retained across sessions.	_id, user, items, total, couponCode, discount
D6	Reviews	MongoDB	Customer reviews and star ratings linked to products and users.	_id, user, product, rating, body, verifiedPurchase, createdAt
D7	Wishlist	MongoDB	Products saved by each user for future reference — one wishlist per user.	_id, user, items[product, addedAt]
D8	UserActivity	MongoDB	Raw behavioural event log: views, searches, cart actions, purchases, recommendation impressions.	_id, user, product, activityType, metadata, timestamp

4. Data Flows

Named data flows between external entities, processes, and data stores.

Flow	From	To	Data	Description
F1	Customer (E1)	P1 Authentication	Email, password or registration form data	User submits login credentials or new account registration
F2	P1 Authentication	Customer (E1)	JWT access token, user profile object	System returns auth token and profile on successful authentication
F3	P1 Authentication	D1 Users	User document (create or update)	New user created or existing user record updated in Users store
F4	Admin (E2)	P2 Product Management	Product data, category data, image files	Admin creates or modifies product listings and categories
F5	P2 Product Management	D2 Products	Product document with Cloudinary image URLs	Persist product data after Cloudinary upload completes
F6	Customer (E1)	P3 Order Processing	Cart ID, shipping address, coupon code	Customer initiates checkout
F7	P3 Order Processing	D3 Orders	New order document with pending status	Order written to store after cart validation and stock reservation
F8	P3 Order Processing	P4 Payment Processing	Order ID, amount, payment method, phone / email	Checkout triggers payment initiation
F9	P4 Payment Processing	Payment Gateway (E3)	Payment request: amount, reference, redirect URLs	System sends payment intent to selected gateway
F10	Payment Gateway (E3)	P4 Payment Processing	IPN / webhook: status, transaction ID, raw	Gateway notifies system of payment success or failure

			payload	
F11	P4 Payment Processing	D4 Payments	Payment record with final status	Write payment outcome and raw gateway response for audit trail
F12	P4 Payment Processing	D3 Orders	Updated paymentStatus and order status	Order record updated following confirmed or failed payment
F13	Customer (E1)	P7 Activity Tracking	Event type, product ID, metadata	User interaction events (views, searches, cart actions) sent to tracker
F14	P7 Activity Tracking	D8 UserActivity	Activity log entry	Raw behavioural event persisted for analytics and AI training
F15	P7 Activity Tracking	AI Service (E4)	Aggregated user behaviour summary	Behaviour data forwarded to AI service for recommendation scoring
F16	AI Service (E4)	P5 Recommendation Engine	Ranked product ID list with recommendation type	AI service returns recommendation results for the requesting user
F17	P5 Recommendation Engine	Customer (E1)	Enriched and ranked product list	Personalised or trending products delivered to the customer UI
F18	Customer (E1)	P6 Review and Wishlist	Review content (rating, body) or wishlist action	User submits a review or adds/removes a wishlist item
F19	P6 Review and Wishlist	D6 Reviews / D7 Wishlist	Review document or updated wishlist array	Review or wishlist state persisted to respective data store
F20	P6 Review and Wishlist	D2 Products	Updated averageRating and reviewCount	Product aggregate rating fields recalculated after each new review